

MER-Live: An Interactive Browser Demo of Prosody-Driven Multimodal Emotion Recognition

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Abstract

MER-Live is a real-time multimodal emotion recognition demo in which speech is the dominant signal, supported by video and lexical cues. The acoustic branch uses a novel lightweight masked auto-encoder SER model (MSMC) on log-mel spectrograms, while the text branch acts only as an auxiliary tie-breaker rather than a keyword trigger. Users speak to the camera and watch emotion estimates update four times per second. They can hot-swap ten speaker-holdout checkpoints, record a five-second utterance for IEMOCAP-style logit averaging, or upload audio files for side-by-side comparison with an open SER baseline. The 6.77M-parameter audio model, exported to TensorRT FP16, serves a batch of five in 0.24 ms on one H200, giving a 14x speed-up over PyTorch FP32. On 10-fold IEMOCAP, the deployed audio-only MSMC checkpoint used in MER-Live reaches 74.03% WA and 66.39% UA. Multimodal performance exceeds 83%.

Index Terms: multimodal emotion recognition, prosody, real-time deployment, interactive demonstration, IEMOCAP

1. Introduction

Most published speech-emotion recognition (SER) systems are evaluated offline, and average segment-level logits across an utterance whose boundaries are known in a priori[1]. Live deployment is harder, requiring a sliding window, not an oracle boundary, and a real-time inference latency target. Combining modalities such as video and text means fusing different rates and scales of mode predictions, in real-time.

MER-Live is the deployment-grade companion to a recently developed lightweight SER architecture, having three main characteristics: (i) a lightweight 6.77 M parameter SER backbone that deliver sub-millisecond latency on a commodity GPU through TensorRT FP16 [2]; (ii) a transparent confidence-driven late-fusion mechanism combining vision, voice, and text in a context-sensitive fashion. (iii) high performance in both real-time and offline evaluations.

The highly efficient implementation allows the resulting system to be small enough to run end-to-end in a browser tab. While not trained on a single individual’s voice, the implementation allows hot-swapping of parameter weights derived from each of the 10-fold IEMOCAP hold-out speaker sets [3]. This aligns with the Interspeech 2026 “Speaking Together” theme: every audience member can speak, in any language, and immediately see how the model responds to their voice.

2. Operation of the Demonstrator

The demon runs in Chrome browser. It displays an emoji and four emotion bars (Happy/Sad/Angry/Neutral) that update every 250 ms. A modality-weight inspector panel on the right shows in real time which mode (face, voice, or text) is currently driving the decision. There are five ways to use the demo:

- **Real-time interaction:** Facing the camera and speaking into the microphone. For example a user can say “*I had a great day*” cheerfully, then flatly, then angrily while maintaining the same facial expression. Because the lexical content is fixed, and the video is similar, only the acoustic branch varies. The audio bars swing from *hap* to *neu* to *ang* while the modalities remain roughly constant.
- **Hot-swap hold-out speaker:** A dropdown loads any of ten fold-specific checkpoints (the speaker held out at training time). Parameter update takes approximately 150 ms because the corresponding TensorRT engine is cached on disk.
- **Record 5 s:** This captures a 5 s clip, runs sliding-window logit averaging and returns one verdict. This is the closest the demo gets to the offline IEMOCAP evaluation protocol.
- **Demo wav:** Streams a real held-out IEMOCAP utterance through both the laptop speakers and the WebSocket. The live UI reacts to the wav file input.
- **Audio File Test:** Drag-and-drop .wav/.mp3 or .flac files for analysis. The result table shows our SER model and the open SER baseline side by side, and per-call latency.

3. Why It Is Acoustic, Not Lexical

A common misinterpretation of any multimodal demo that involves ASR is that it is secretly a keyword classifier. MER-Live is explicitly designed so the audience can verify that on the spot: the audio model never sees text, phonemes or word boundaries. Its only input is a 128×300 log-mel spectrogram of the most recent three seconds of audio. The features that drive its decision are therefore strictly **non-lexical**, i.e. prosody.

The backbone is a Masked Auto-Encoder [4] pretrained on roughly 17.7k IEMOCAP segments without labels, then fine-tuned with a four-class head on the standard speaker-independent fold split. Self-supervised pretraining on raw spectrograms makes the encoder a prosodic feature extractor first, and an emotion classifier only via a small linear head afterwards. This is the property that makes the “same-words-different-tone” interaction mentioned in Section 2 work.

The text branch is more of a **tie-breaker** than a driver. It receives ASR transcripts from a native SpeechRecognition

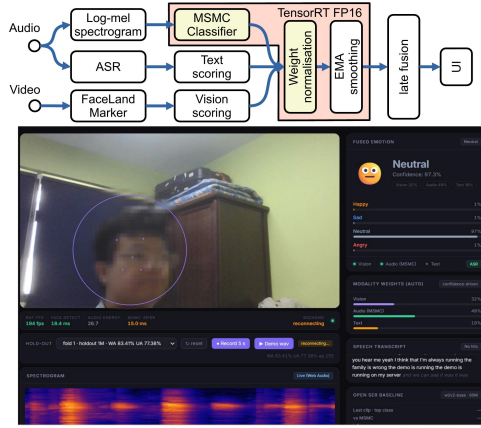


Figure 1: End-to-end MER-Live pipeline (top) with UI screenshot (below).

Table 1: Top-1 (%) of the deployed fusion rule on Ses01M with synthetic vision and text probability streams. Parenthesised values are vision+text only, with no audio.

vis \ txt	55 %	70 %	85 %	100 %
55 %	62.4	71.0	76.1	81.7
70 %	70.6	76.5 (70.4)	80.2	87.6
85 %	73.3	82.2	87.1 (84.9)	92.3
100 %	85.4	89.2	93.0	97.9

API and emits a soft probability vector that contributes via the same confidence-weighted fusion rule used by every modality (sect. 4). When prosody is informative, the audio weight dominates and other modalities are overridden; when the user is silent the audio weight collapses to zero and the headline emoji follows the face and text. The fusion rule is symmetric.

4. Confidence-Driven Late Fusion

Each modality m is given a weight to suppress its vote when it is confident the user is *not* expressing emotion via that modality: $w_m = 2 \max(0, (1 - p_m^{\text{neu}}) - 0.15)$. Weights are renormalised, summed, weighted, then smoothed by a display-side EMA ($\alpha = 0.20$) with a 0.10 hysteresis on the dominant-class label. Since IEMOCAP video is not face/text-annotated, we cannot measure multimodal accuracy directly. Instead we can fix the audio probabilities and synthesise vision and text probability vectors at controlled per-branch accuracy levels, then run them through the JavaScript late fusion rule. Table 1 shows that audio improves a vision+text baseline by 6 to 9%, achieving a performance of 70 to 85%.

5. Real-Time Deployment Stack

The MSMC encoder and classifier are exported to ONNX (opset 17) and compiled offline to a TensorRT 10.5 FP16 engine [2], one per fold (build time approximately 80 s; 15 MB on disk). The mel-spectrogram step remains in PyTorch (it does not trace cleanly to ONNX), but more than 90 % of the original latency was the encoder graph, which is now native TensorRT.

The PyTorch model remains as a fallback when a fold-specific TensorRT engine is unavailable, while the frontend reports the active backend via a runtime metadata endpoint. Table 2 details the latencies.

For each utterance-level inference, the same audio clip

Table 2: Inference latency on one NVIDIA H200, batch 5 (5-shift TTA), warm cache, mean of 1k calls.

Backend	Latency	Speed-up
PyTorch FP32 (eager)	3.36 ms	1.0×
TensorRT 10.5 FP16	0.24 ms	14.0×
Numerical drift (FP16 vs FP32): max abs logit error ≈ 0.001		

is also evaluated using a public HuggingFace wav2vec 2.0 SER baseline [5, 6] with the same four-class label set. This 95M-parameter baseline provides an open reference against our 6.77M-parameter model, reporting the predicted class, agreement, and latency in the UI. It requires approximately 840 ms for cold start and 6 ms per warm inference.

6. Conclusion

MER-Live demonstrates an end-to-end multimodal emotion recognition pipeline, where the spoken analysis branch demonstrates a novel lightweight data-driven prosody-based recogniser¹, and is augmented by text (from ASR) and image-based modalities. Despite exceeding real-time performance, the system has several limitations: (a) The real-time mode accuracy has a ceiling near to 60 % due to the fixed three-second receptive field. Achieving the full 83 % fold-1 performance requires a longer inference, hence the *Record 5 s* mode. (b) The text branch requires Chrome or Edge browsers, but will still operate well (with no ASR) on other browsers. (c) The current build serves over plain WebSocket. `getUserMedia` therefore only works on localhost or behind a TLS terminator.

7. Use of Generative AI Disclosure

The authors used generative AI tools for language polishing and grammar checking only. All scientific content, system design, experiments, analysis, and conclusions were created and verified by the authors.

8. References

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¹Source code, and runtime package will be on GitHub.